FORENSICS

3 - Windows, Memory

STT

!! nos4a2
@sluzypt

AFSC19.EXE
@afsc19

ROAD MAP

~~1. Foundations~~

~~2. Disks, registries, network~~

3. Windows, memory

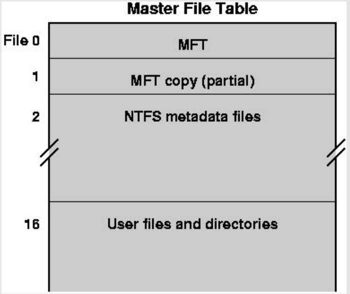
4. ???

Windows

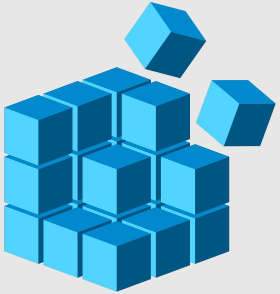

MICROSOFT
Microsoft Windows
Version 1.01
Microsoft
Windows
Version 3.0
MICROSOFT.
WINDOWS NT.
Version 3.1
Microsoft
Microsoft
Windows 95
Microsoft
Windows 98
Microsoft
Windows xp
Windows Vista
Start
skype
NETFLIX
amazon
Windows 10
Windows 11

Windows - Stores (almost) everything!

Event Logs



MFT



Registry

Prefetch

•••

Windows Registry - Locations

C:\Users\<user>\NTUSER.dat

C:\Windows\System32\config

Folder/predefined key	Description
HKEY_CURRENT_USER	Contains the root of the configuration information for the user who is currently logged on. The user's folders, screen colors, and Control Panel settings are stored here. This information is associated with the user's profile. This key is sometimes abbreviated as *HKCU*.
HKEY_USERS	Contains all the actively loaded user profiles on the computer. HKEY_CURRENT_USER is a subkey of HKEY_USERS. HKEY_USERS is sometimes abbreviated as *HKU*.
HKEY_LOCAL_MACHINE	Contains configuration information particular to the computer (for any user). This key is sometimes abbreviated as *HKLM*.
HKEY_CLASSES_ROOT	Is a subkey of `HKEY_LOCAL_MACHINE\Software`. The information that is stored here makes sure that the correct program opens when you open a file by using Windows Explorer. This key is sometimes abbreviated as *HKCR*. Starting with Windows 2000, this information is stored under both the HKEY_LOCAL_MACHINE and HKEY_CURRENT_USER keys. The `HKEY_LOCAL_MACHINE\Software\Classes` key contains default settings that can apply to all users on the local computer. The `HKEY_CURRENT_USER\Software\Classes` key contains settings that override the default settings and apply only to the interactive user. The HKEY_CLASSES_ROOT key provides a view of the registry that merges the information from these two sources. HKEY_CLASSES_ROOT also provides this merged view for programs that are designed for earlier versions of Windows. To change the settings for the interactive user, changes must be made under `HKEY_CURRENT_USER\Software\Classes` instead of under HKEY_CLASSES_ROOT. To change the default settings, changes must be made under `HKEY_LOCAL_MACHINE\Software\Classes`. If you write values to a key under HKEY_CLASSES_ROOT, the system stores the information under `HKEY_LOCAL_MACHINE\Software\Classes`. If you write values to a key under HKEY_CLASSES_ROOT, and the key already exists under `HKEY_CURRENT_USER\Software\Classes`, the system will store the information there instead of under `HKEY_LOCAL_MACHINE\Software\Classes`.
HKEY_CURRENT_CONFIG	Contains information about the hardware profile that is used by the local computer at system startup.

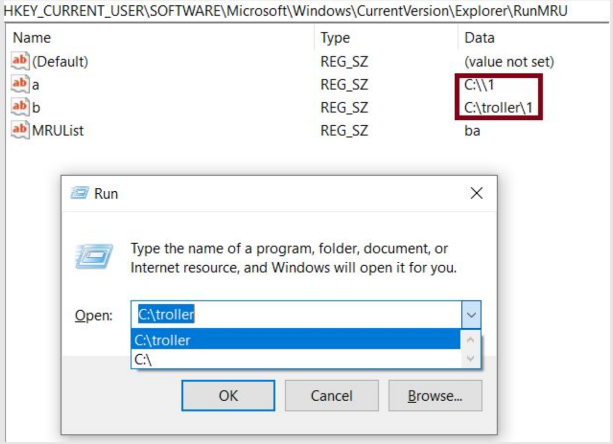
Windows Registry - Execution evidence

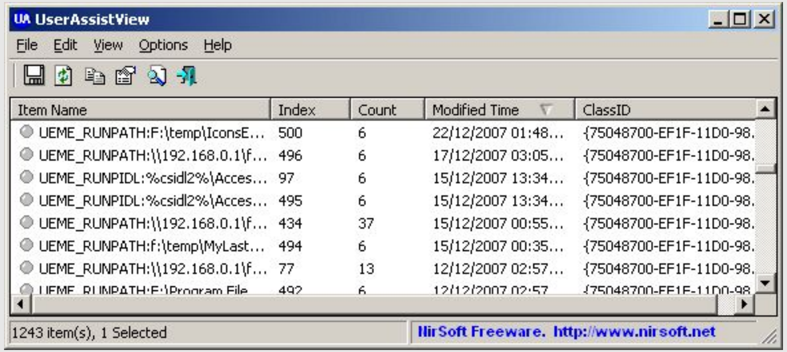
UserAssist

- HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist

RunMRU

- HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU





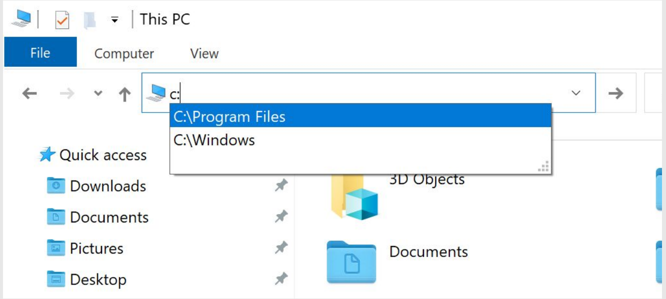
Windows Registry - User activity

Shellbags

- HKCU\Software\Microsoft\Windows\Shell\Bags

Typed paths (in Windows Explorer)

- HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths



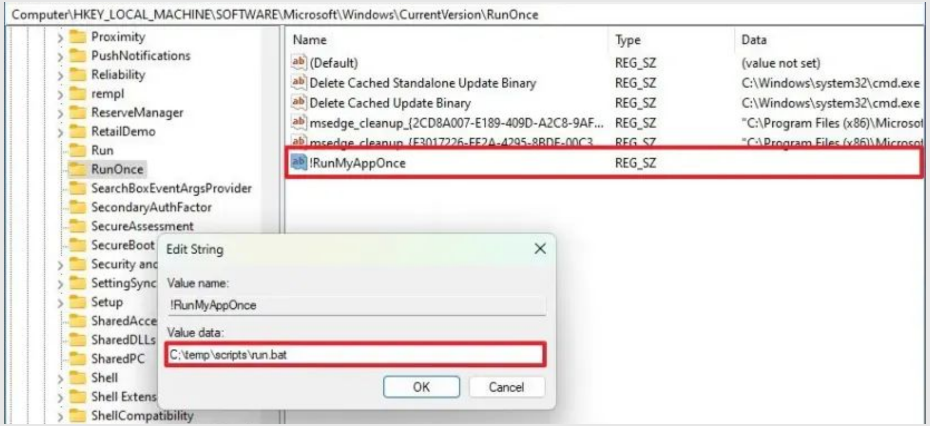
Windows Registry - Startup programs

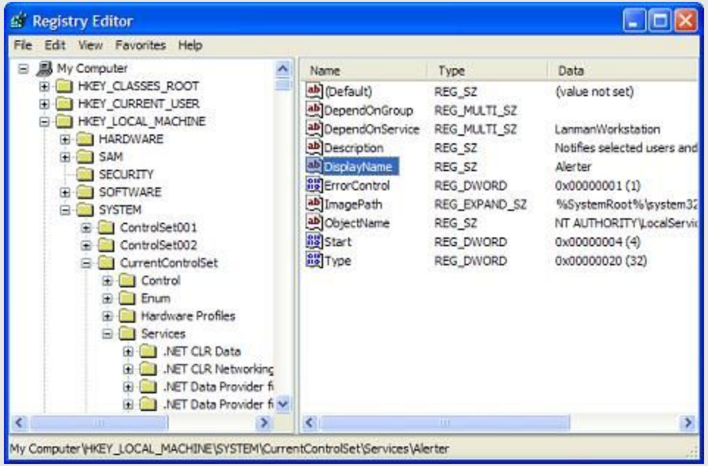
Run/RunOnce

- HKLM\Software\Microsoft\Windows\CurrentVersion\Run[Once]

Services

- HKLM\SYSTEM\CurrentControlSet\Services





Windows Registry - Other informations

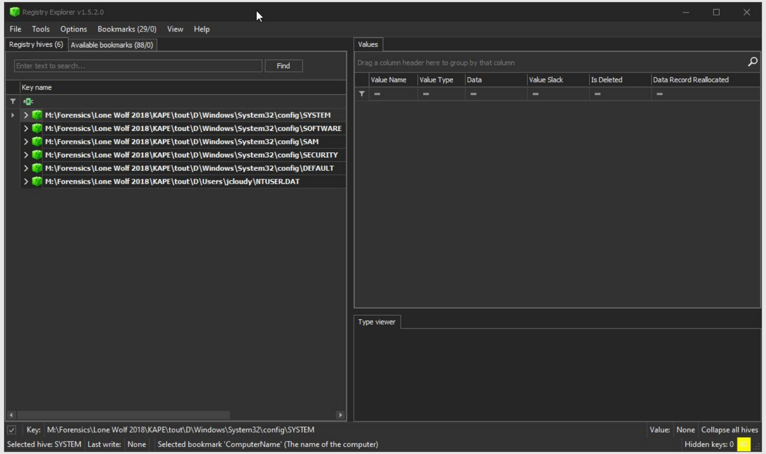
- USB & External Device History (USBSTOR)
- Network Configuration & History (NetworkList)
- Mounted Devices (MountedDevices)
- Windows Time Zone Information
- Audit Policy Settings
- System Information & OS Versioning
- Computer Name & Domain Membership
- Background Activity Moderator (BAM/DAM)

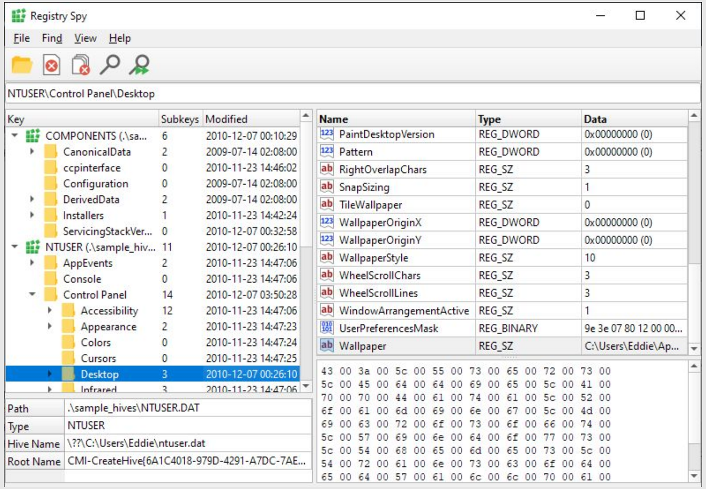
And so on ....

Windows Registry - Other informations

Windows Registry - Tools

- Registry Explorer (Windows - [Eric Zimmerman’s Tools](#))
- [Registry-Spy](#)





Windows Event Logs

Security

- Authentication & Logon Activity
- Account Management
- Audit Policy & Log Tampering
- Privilege Escalation

Application

- Software Crashes/Faults
- Software Installs
- Database Engine (ESENT) Events
- App-Specific Audits

Windows Event Logs

System

- Service Installation
- System Health
- Time Changes
- Network/IP Events

Check what event IDs you're looking for, using a list.

Tools:

- Windows Event Viewer itself
- omerbenamram/evtx
- VS Code EVTX extension
- Python-evtx

Windows Security Log Event ID 4713

4713: Kerberos policy was changed

On this page

- Description of this event
- Field level details
- Examples

Operating Systems	Windows 2008 R2 and 7 Windows 2012 R2 and 8.1 Windows 2016 and 10 Windows Server 2019 and 2022
Category • Subcategory	Policy Change • Authentication Policy Change
Type	Success
Corresponding events in Windows 2003 and before	617

Windows logs 4713 when it detects a change to the the domain's Kerberos policy. Kerberos policy is defined in GPOs linked to the root of the domain under Computer Configuration\Windows Settings\Security Settings\Account Policy\Kerberos Policy.

Unfortunately the Subject fields don't identify who actually changed the policy because Kerberos policy isn't directly configured by administrators. Instead it is edited in a group policy object which then gets applied to the computer. Therefore this event always shows the local computer as the one who changed the policy since the computer is the security principal under which gpupdate runs.

Subject:

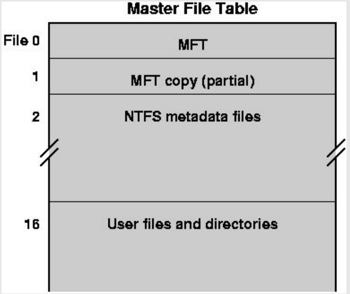
The ID and logon session of the user that changed the policy - always the local system - see note above.

- Security ID: The SID of the account.
- Account Name: The account logon name.
- Account Domain: The domain or - in the case of local accounts - computer name.
- Logon ID is a semi-unique (unique between reboots) number that identifies the logon session. Logon ID allows you to correlate backwards to the logon event (4624) as well as with other events logged during the same logon session.

Lookup for details!

Windows - Tools for the rest

- [Eric Zimmerman's tools]
- [NirSoft's tools]
- [San's tool curation]



MFT
(Alternate data streams)

Prefetch
(Prefetch of files ran)

...

Memory Dumps

Memory Dumps - Information to extract

- Running processes
- Command history
- Files
- Loaded kernel modules
- Network information

…

Memory Dumps - Tools

- Volatility 2 (docker run)
- Volatility 3
- MemProcFS

Why 2 versions of volatility?

Updated to Python3, lost some modules

- imageinfo
- connections, connscan, sockets, sockscan
- deskscan, sessions, atomscan, userhandles
- apihooks
- iehistory
- screenshot
- notepad
- clipboard
- cmdscan, consoles

Volatility - symbols

- Finding banners

```
❯ python3 volatility3/vol.py -f memory.elf banners
Offset          Banner
0x156000e0   Linux version 6.12.73+deb13-amd64 (debian-kernel@lists.debian.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-19)
14.2.0, GNU ld (GNU Binutils for Debian) 2.44) # SMP PREEMPT_DYNAMIC Debian 6.12.73-1 (2026-02-17)
```

- Finding the symbols
  Abyss-W4tcher/volatility3-symbols
- Extracting them

1. Download the debug-symbol vmlinux
2. Extract using dwarf2json

```
❯ dwarf2json linux --elf ./debian-debug/boot/vmlinux-4.4.0-137-generic > output.json
```

Dynamic Analysis

Remote

- Any.run
- tria.ge
- Joe Sandbox
- Cuckoo Sandbox

Local (Inside a VM)

- API Monitor
- Windows SysInternals

Other Tools

- mesquidar/ForensicsTools
- cugu/awesome-forensics